BT4222 AY25/26 S2: GROUP 1 FINAL PROJECT PRESENTATION

DUNNHUMBY RECOMMENDER SYSTEM

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BUSINESS ISSUE

- Challenges:
  - Habitual inertia and latent demand
  - Paradox of choice

- Goals:
  - Basket expansion
  - Encourage product discovery
  - Increase cross-category shopping
  - Personalised and relevant recommendations

DATASET OVERVIEW

DATASET

- Integration of transaction records, household demographics, and product attributes
  - 2,595,732 transaction records
  - 801 households
  - 92,353 unique products
- Transaction spans 711 days
- Key columns:
  - sales_value, retail_disc, coupon_disc, coupon_match_disc
  - age_desc, income_desc, marital_status, homeowner_desc, hh_comp_desc
  - department, brand, commodity_desc, sub_commodity_desc

DATA CLEANING

- NAs
  - 1,168,429 entries had missing demographic information
  - Dropped 7,839 missing product descriptions
  - Dropped 120,328 missing household composition descriptions

  → Dropped 1,296,596 rows where essential features were missing

- Erroneous transactions
  - Dropped 16 positive retail_disc transactions
  - Dropped 2,840 zero-quantity transactions (incl. 12 positive retail_disc)
  - Dropped 12,916 $0 shelf-price products

  → Dropped 14,208 rows with erroneous transactions

- Post-cleaning: 1,283,376 data entries left

FEATURE ENGINEERING

MAIN GOAL

- Capture user demographic-specific details
- Allow the model to learn user-item interactions
- Enhance behavioural generalisation
- Capture multi-dimensional sensitivity
- Mitigate data sparsity in the long tail

USER FEATURES

REGULARITY

Days between purchases
The higher the value, greater the variance and irregularity of purchases

COMMODITY ENTROPY

Focuses on how similar the items bought are
Low entropy: Products are highly differentiated
High Entropy: Products are nearly identical

DECAY FREQUENCY

Higher weight is assigned to recent transactions
Capture gradual preference shifts over time
Incorporate recency into user representation

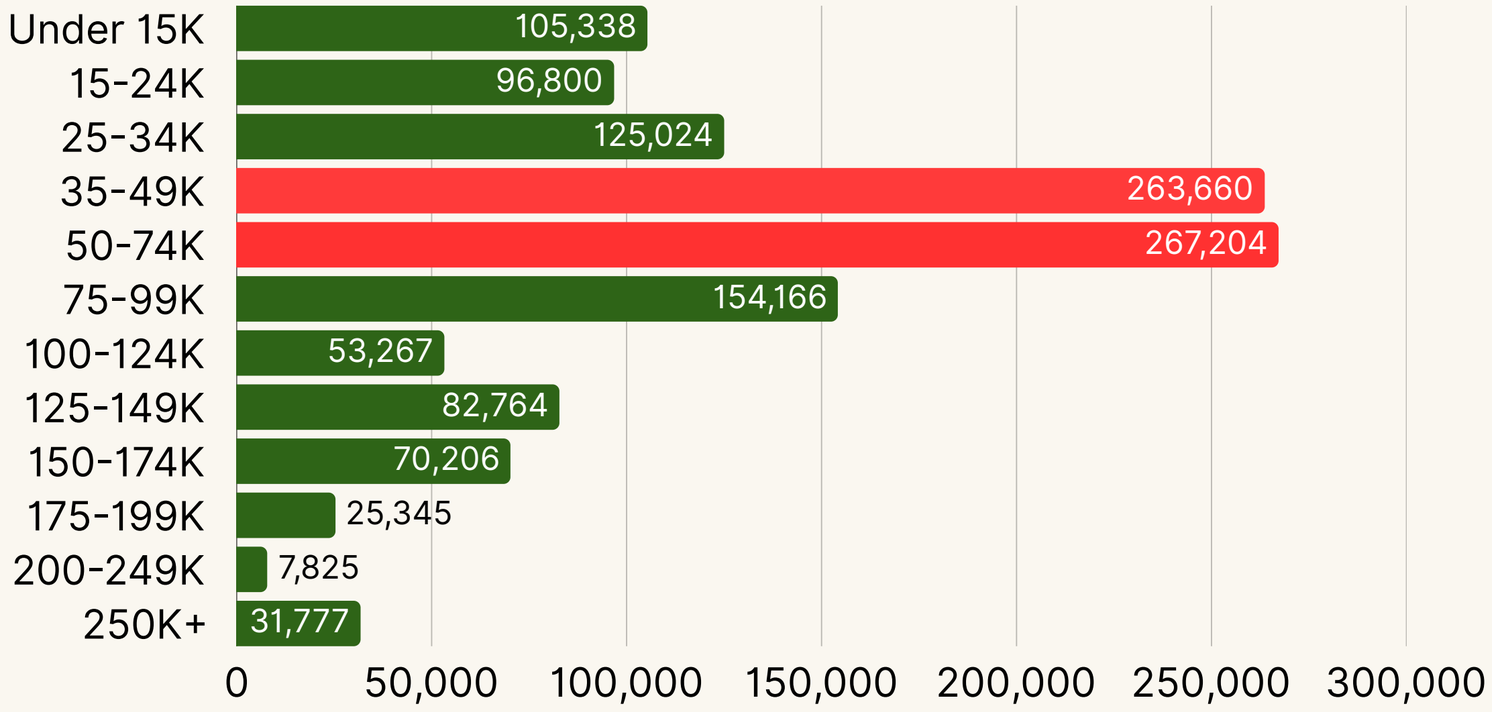
USER PRIVATE LABEL LOYALTY

Captures the proportion of private label purchases per household
Differentiate price-sensitive value shoppers and brand-conscious consumers

USER FEATURES

INCOME DESCRIPTION

Ability to consider affordability on top of preferences, switching between prioritising quality and value



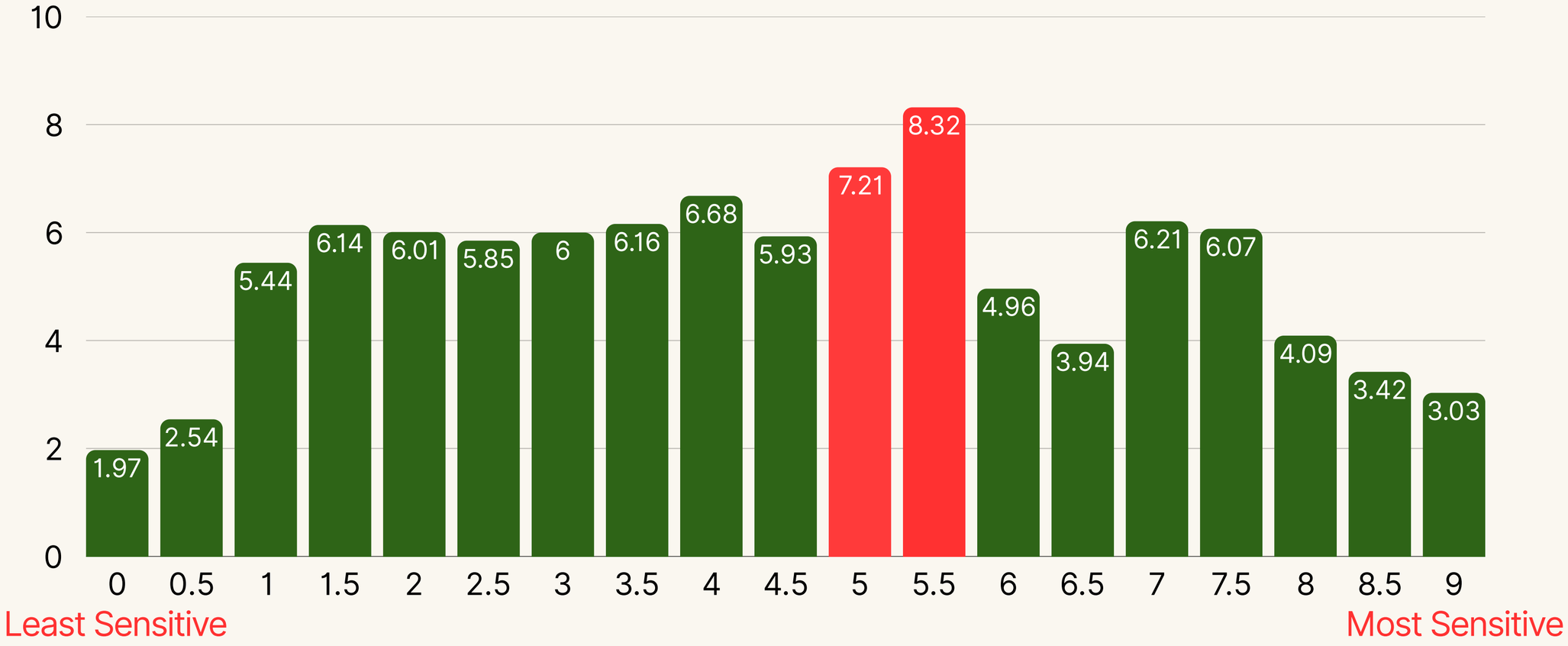
Over 536,000 transactions are between 35k and 74k brackets

Great middle-market dominance

USER FEATURES

PRICE SENSITIVITY DESCRIPTION

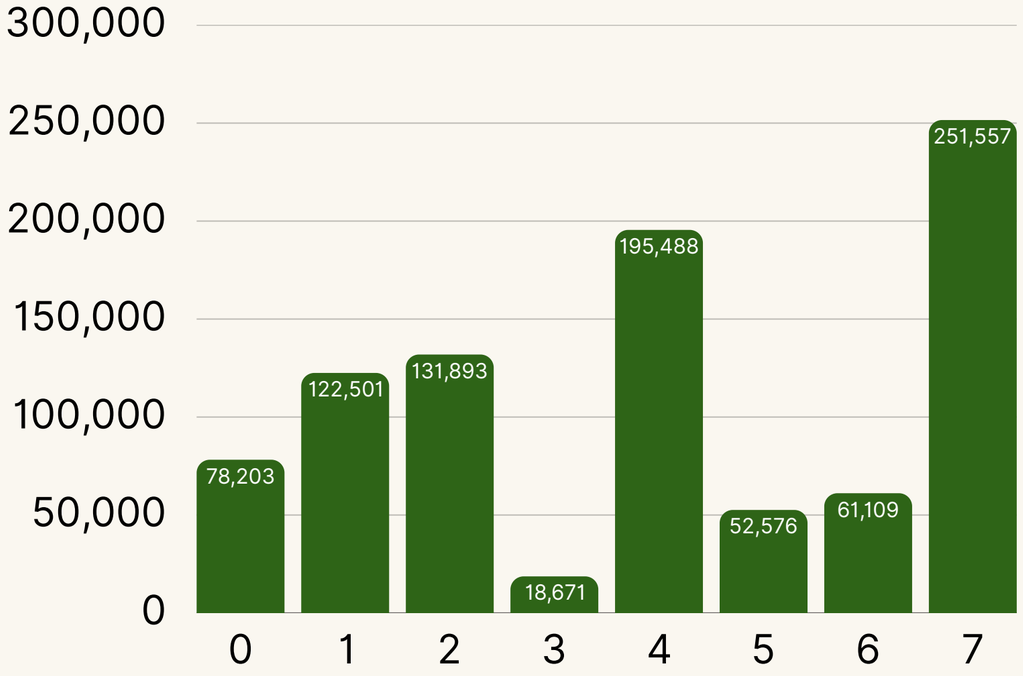
Distribution is relatively normal, spreading equally across difference price elasticity levels



USER FEATURES

LIFE STAGE

Categorise customers into distinct groups to capture patterns behind consumption behaviours



0: Young family
1: Older family
2: Others with kids
3: Young single
4: Middle-aged single
5: Retired/Older single
6: Retired/Older couple
7: Couple without kids
8: Others

Derived From:

Age description
Has kids
Number of adults

ITEM FEATURES

CO PURCHASE

Identifies complementary goods
Uncovers relationship between products

AVERAGE BASKET SIZE

Based on each item
Differentiate the type of grocery trip the product is usually bought in

BASKET CENTRALITY

Finds out how many baskets contains the specific product, identifying the popularity

ITEM FIRST PURCHASE

Flags out the first transaction for an item, filtering out cold items

ITEM AVERAGE GAP

Identifies how frequently an item is being bought
Distinguish between high-frequency items and slow-moving products

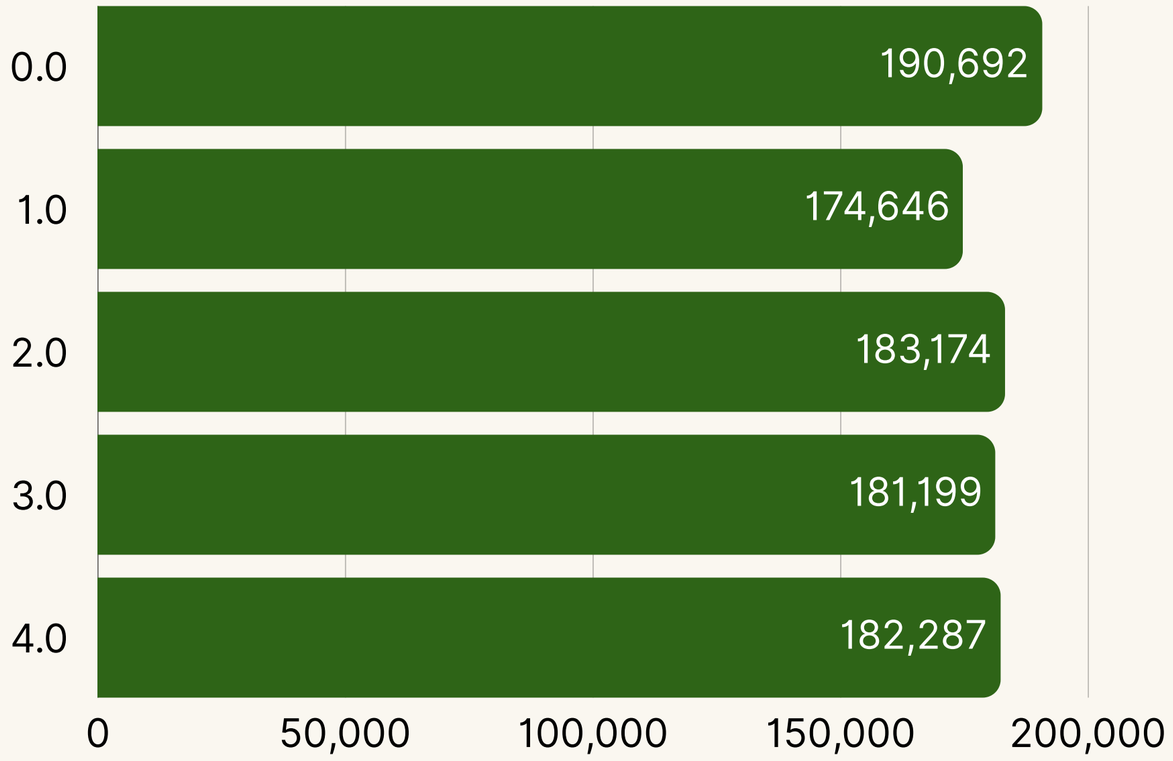
DEPARTMENT

Looks at the department the item belongs to, providing a broad category for items

ITEM FEATURES

ITEM PRICE TIER

Segmenting products into different levels of affordability



Tier 0
$0.99

Tier 1
$1.92

Tier 2
$2.75

Tier 3
$3.97

Tier 3
$299.99

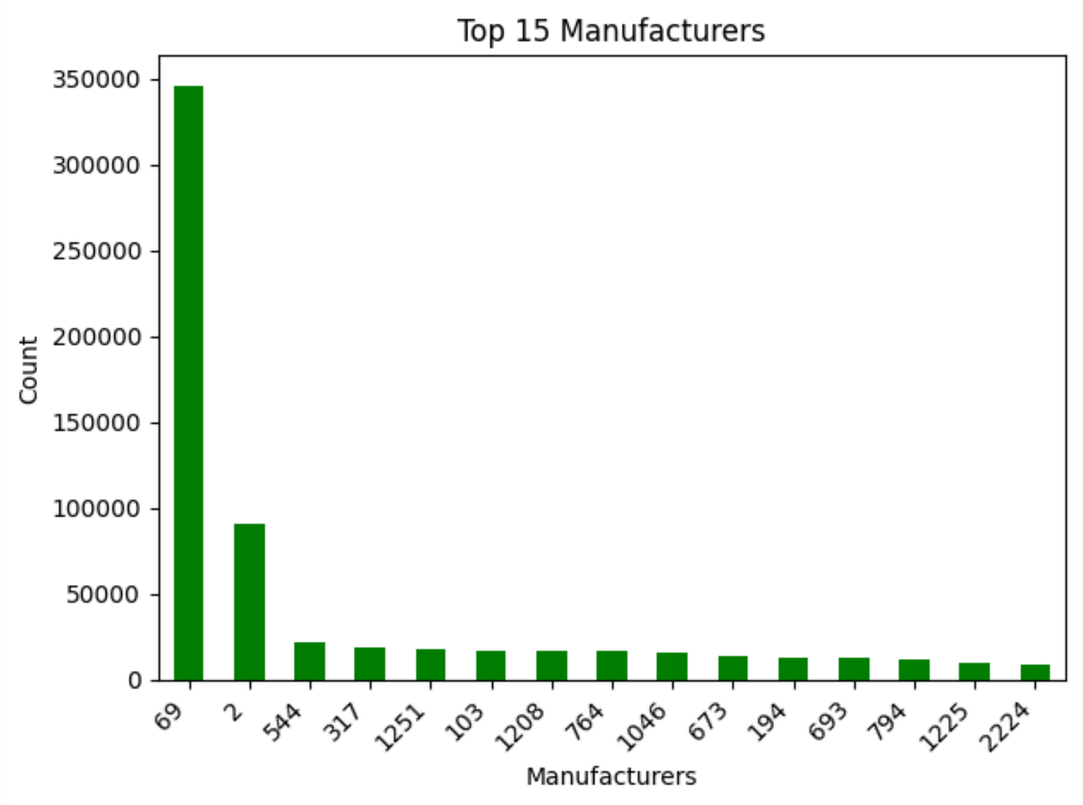
Derived From:

Non member shelf price

ITEM FEATURES

MANUFACTURER

Flag out brand sensitive consumers and ones that prioritise value



Relatively equal distribution, among manufacturers

Manufacturer 69 has 365,514 transactions, likely being the Private House Brand

MODELS & PERFORMANCE

MODEL ARCHITECTURE

MATRIX FACTORISATION

Baseline model

NEURAL COLLABORATIVE FILTERING

Baseline model

USER-FEATURE MODEL

User embeddings replaced with neural network of engineered user features

ITEM-FEATURE MODEL

Item embeddings replace with feature based representations

TWO-TOWER MODEL

User and Item embeddings generated from feature inputs

MULTI -BRANCH TWO TOWER MODEL

Separate pathways for warm and cold items

MODEL ARCHITECTURE

USER-FEATURE MODEL

- Uses Multi-Layer Perceptron to process engineered user features
- Focuses on underlying characteristics of users
- Make predictions for cold users

MODEL ARCHITECTURE

MULTI-BRANCH TWO-TOWER MODEL

- The warm item branch uses both item embeddings and item features, allowing the model to leverage historical interaction data.

- The cold item branch relies only on item features, enabling the model to score items with little or no interaction history.

- Cold item negatives are included to ensure sufficient gradient signal → improves ability to handle cold-start scenarios.

TRAINING METHODS

NEGATIVE SAMPLING

- 4:1 uniform random strategy → Distinguish between hot and cold items

EARLY STOPPING

- Validation AUC does not improve for 3 epochs → Stops training when performance plateaus

EVALUATION METHODS

1-VS-99 EVALUATION

- Ability to rank 1 positive item against 99 random negatives not found in the Train set

COLD USER NEGATIVE SAMPLING

- Samples users with no history → Force model to rely strictly on feature-based decision rules
- 20% of Users artifically converted to cold users

METRICS

AUC

Overall ranking quality across all thresholds

MRR@10

Average of the reciprocal ranks of the first relevant document found within the top 10 results across a set of queries

RECALL@10

Proportion of relevant items in top-10 recommendations

MODEL PERFORMANCE

Model	AUC	Recall@10	MRR@10
MF	0.8682	0.6252	0.3313
NCF	0.8865	0.6821	0.3827
User-Feature	0.9305	0.7299	0.3925
Item-Feature	0.8765	0.6371	0.3376
Two-Tower	0.8492	0.6242	0.3261
MB-Two-Tower	0.8484	0.6251	0.3223

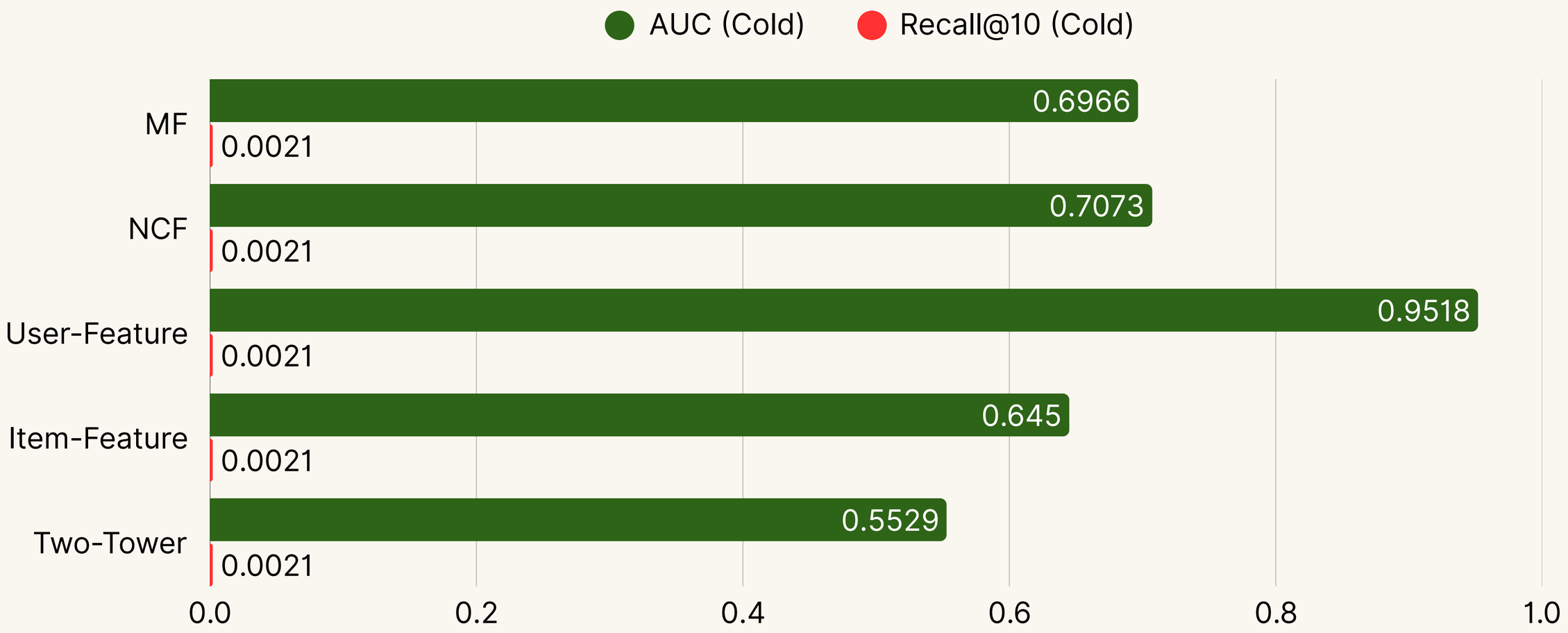
CALIBRATION BY SEGMENT

Model	ECE (Income)	ECE (Life Stage)	ECE (Price Sensitivity)	ECE (Avg Spend)
MF	0.435	0.435	0.435	0.435
NCF	0.42	0.42	0.42	0.42
User-Feature	0.01	0.01	0.01	0.01
Item-Feature	0.405	0.405	0.405	0.405
Two-Tower	0.434	0.434	0.434	0.434
MF	0.435	0.435	0.435	0.435

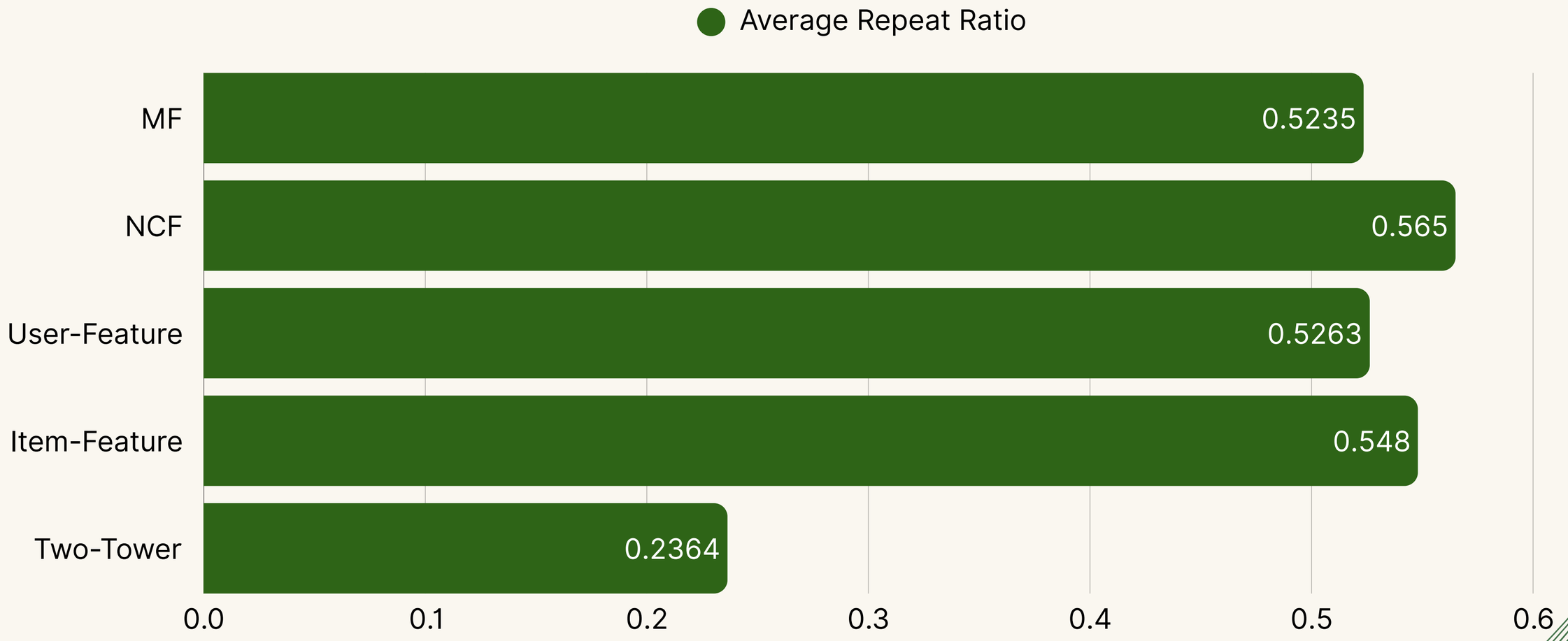
TWO-TOWER ABLATION STUDY

Condition	Recall@10	Drop	Drop %
Full model	0.0382	—	—
No user embedding	0.0382	0	0.00%
No user features	0.041	-0.0027	-7.10%
No user tower	0.041	-0.0027	-7.10%
No item embedding	0.0382	0	0.00%
No item features	0.0112	0.027	70.70%
No item tower	0.0112	0.027	70.70%

COLD-START ANALYSIS



REPEAT VS NEW RECOMMENDATIONS



DEMOGRAPHIC CONSISTENCY

Model	AUC Spread for Income	AUC Spread for Life Stage	AUC Spread for Price Sensitivity
MF	4.50%	1.58%	5.04%
NCF	5.59%	1.63%	6.24%
User-Feature	5.05%	1.27%	4.64%
Item-Feature	2.23%	1.04%	2.28%
Two-Tower	**4.89%**	**1.42%**	**5.44%**

CONCLUSION

Reliable General Recommendation

User-Feature Model

High AUC

Diverse Item Recommendations

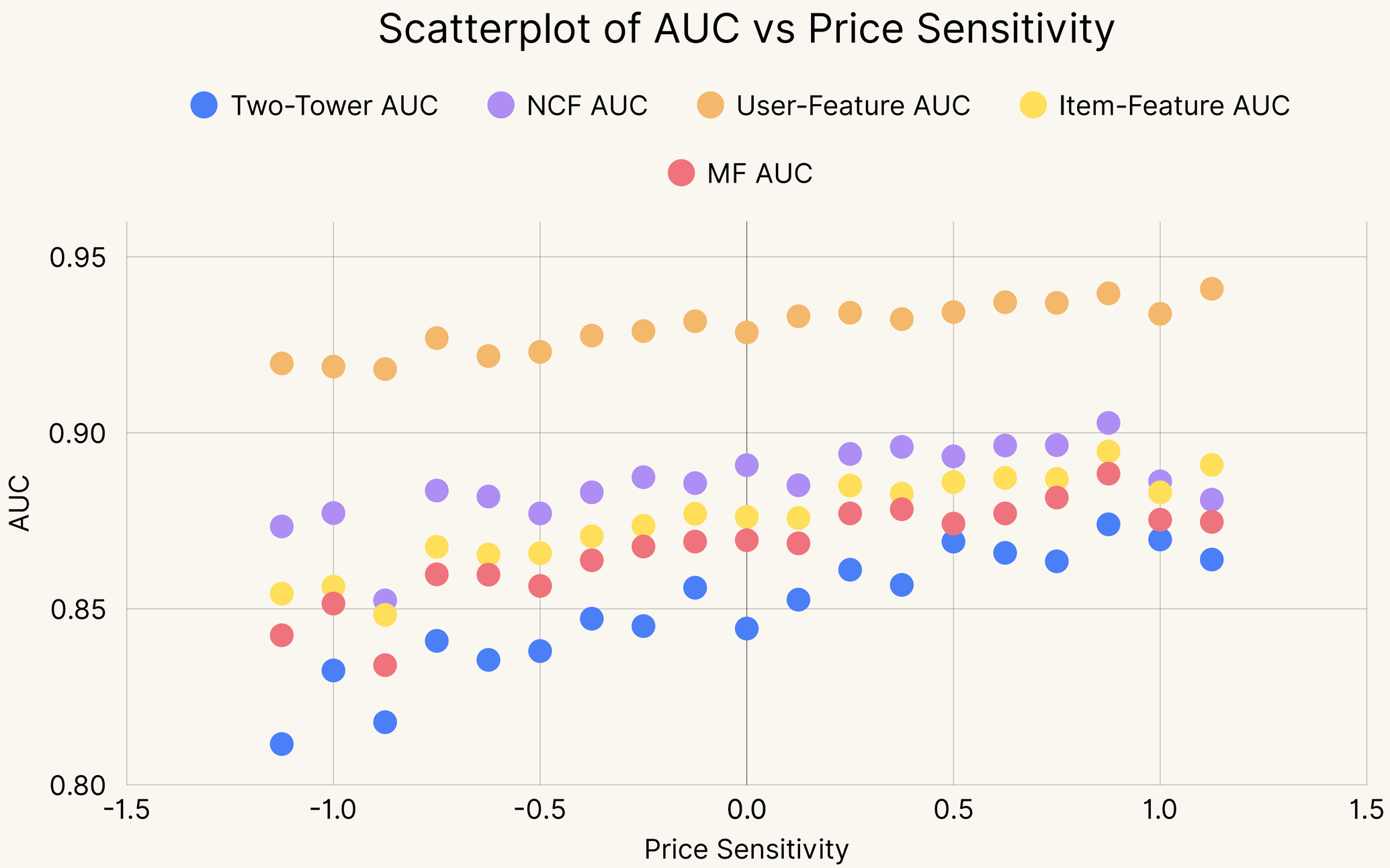
Two Tower Model

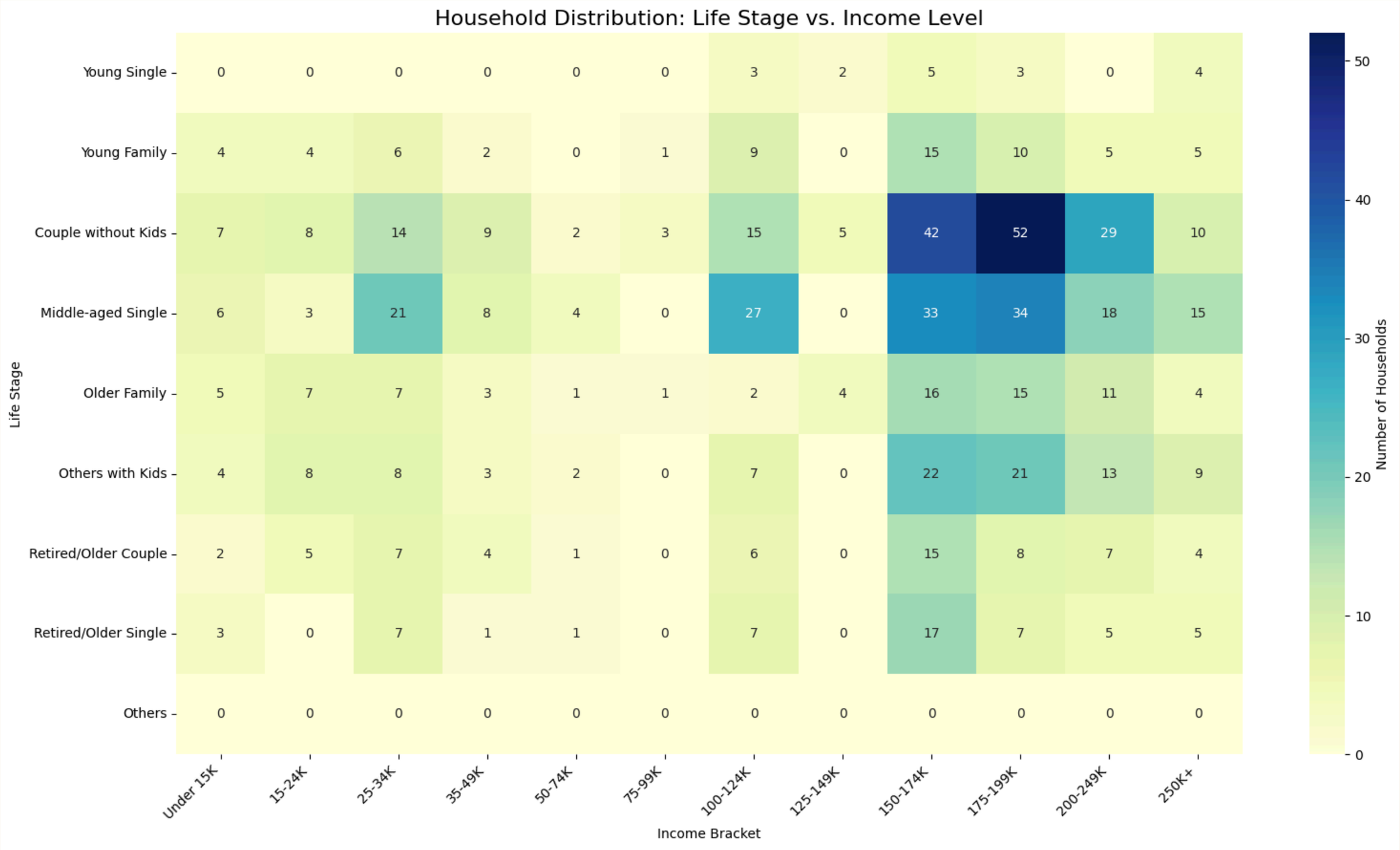
Cold Start Items

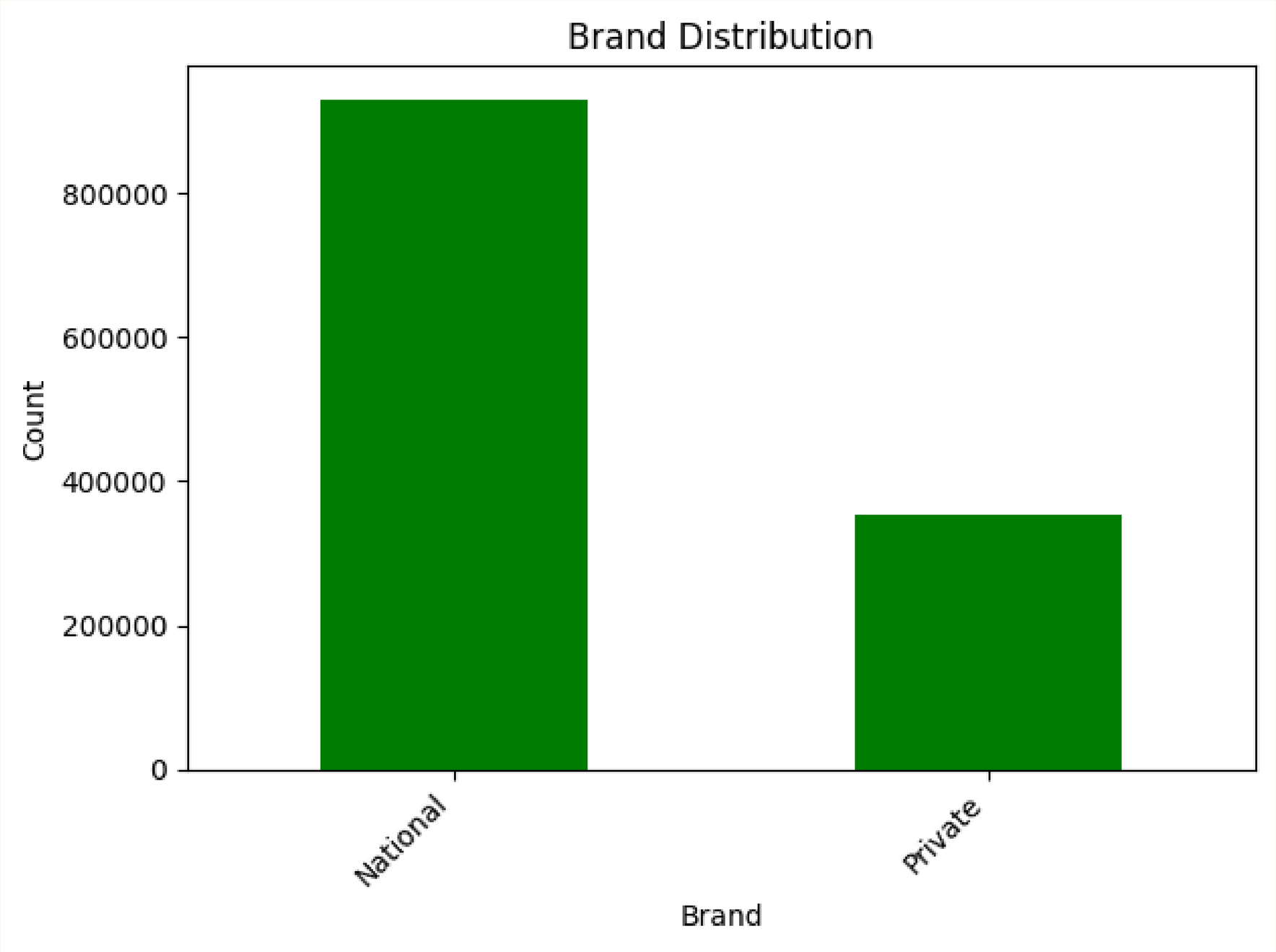
Multi-Branch Two Tower Model

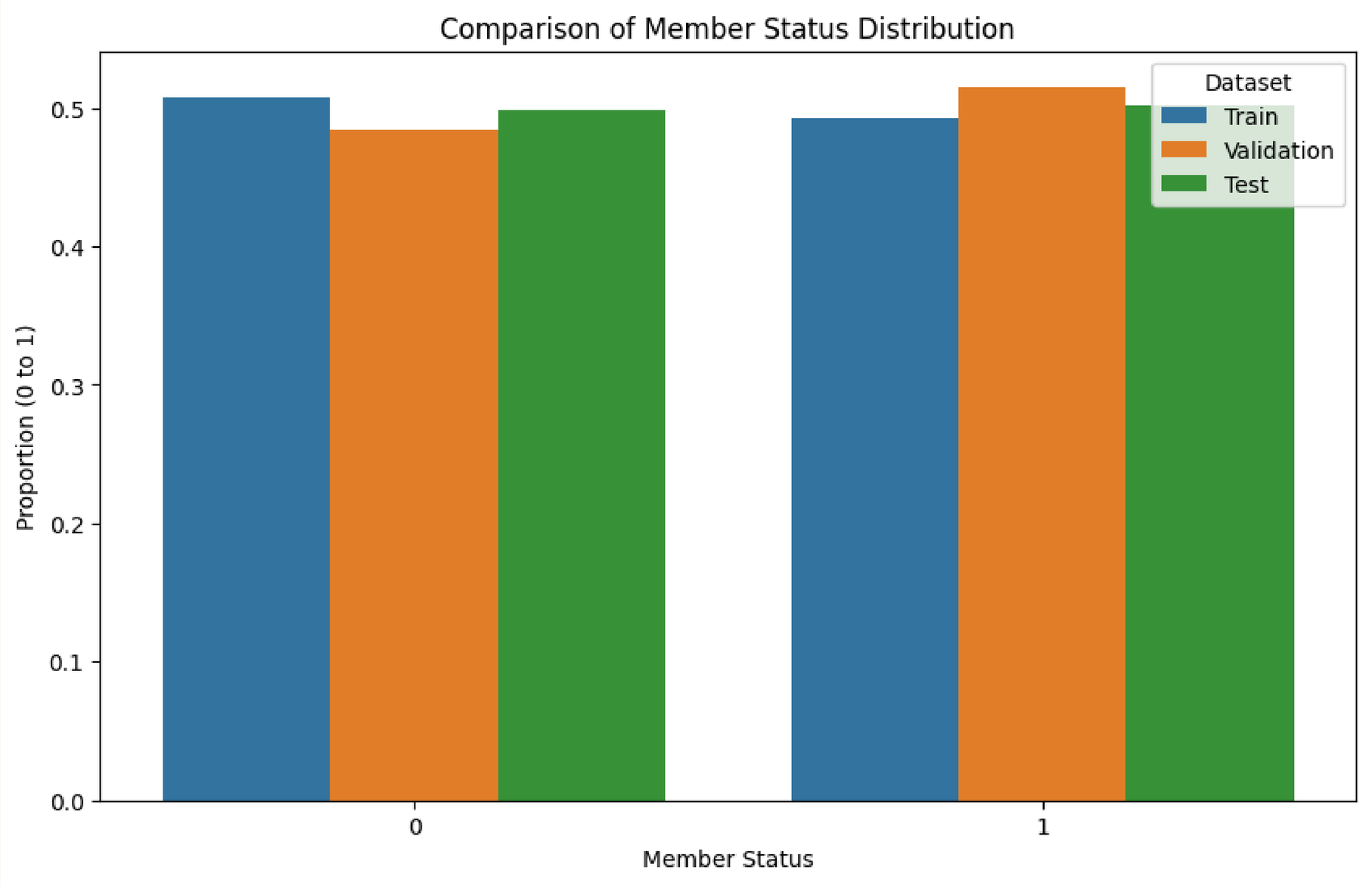
THANK YOU

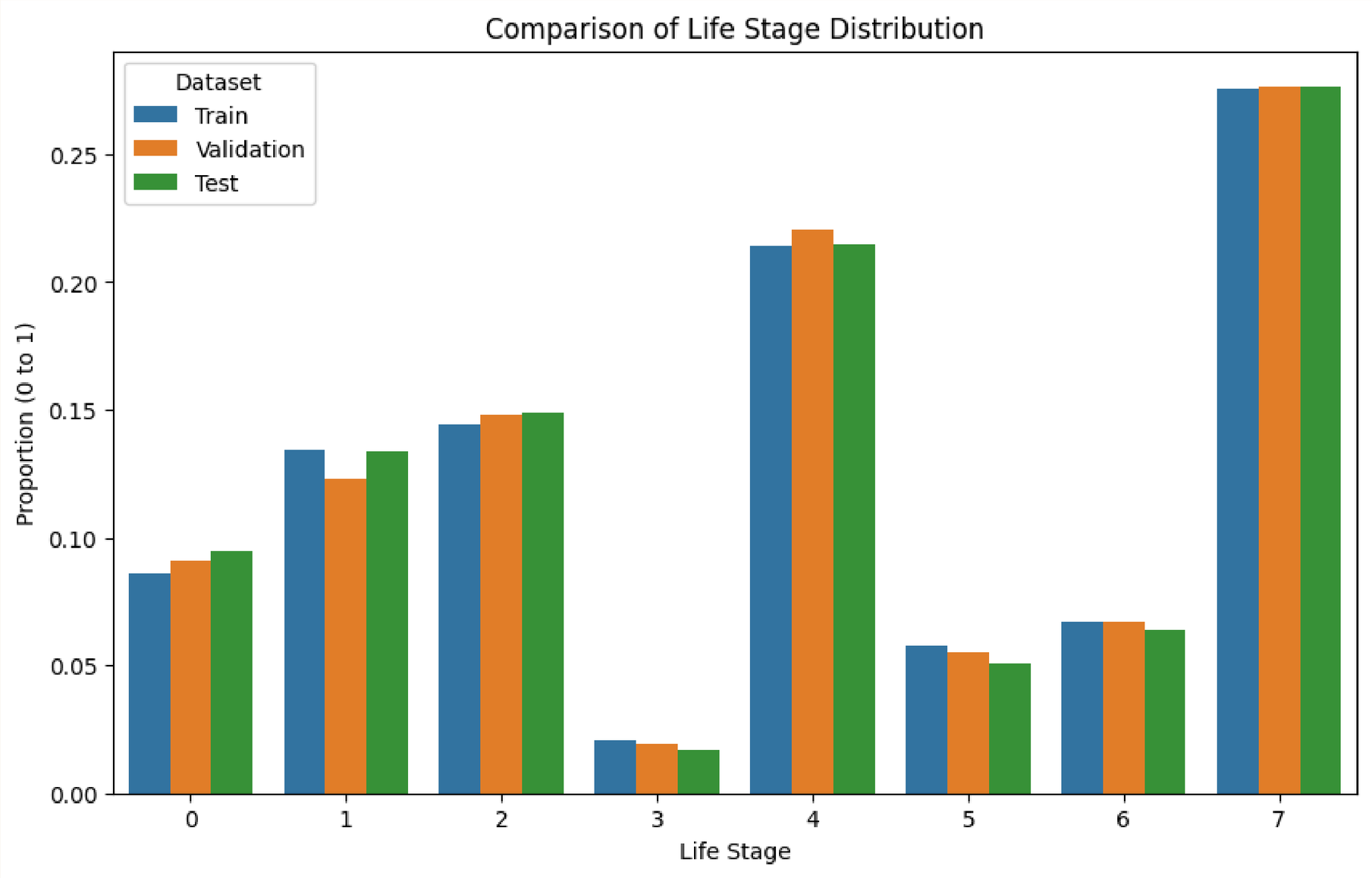
APPENDIX

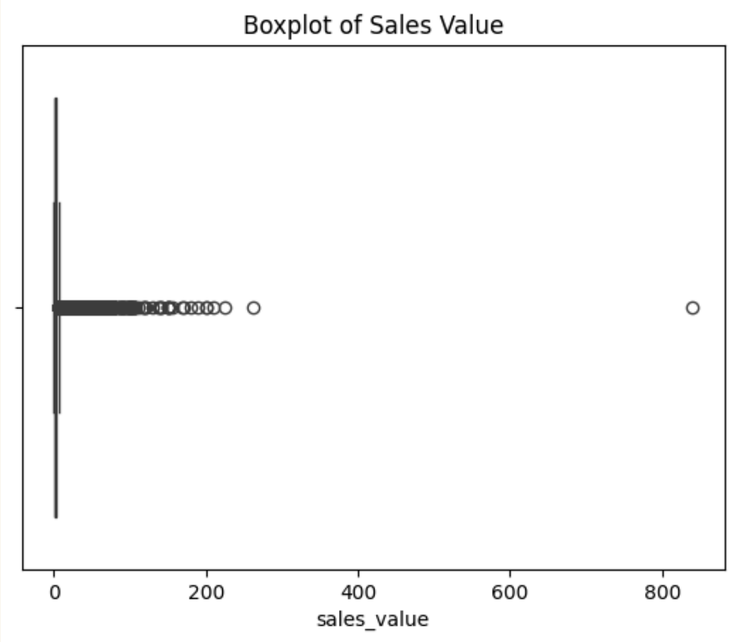

Scatterplot of AUC vs Price Sensitivity
Two-Tower AUC
NCF AUC
User-Feature AUC
Item-Feature AUC
MF AUC
0.95
0.90
0.85
0.80
AUC
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
Price Sensitivity


Household Distribution: Life Stage vs. Income Level
Life Stage
Income Bracket
Number of Households
Young Single 0 0 0 0 0 0 3 2 5 3 0 4
Young Family 4 4 6 2 0 1 9 0 15 10 5 5
Couple without Kids 7 8 14 9 2 3 15 5 42 52 29 10
Middle-aged Single 6 3 21 8 4 0 27 0 33 34 18 15
Older Family 5 7 7 3 1 1 2 4 16 15 11 4
Others with Kids 4 8 8 3 2 0 7 0 22 21 13 9
Retired/Older Couple 2 5 7 4 1 0 6 0 15 8 7 4
Retired/Older Single 3 0 7 1 1 0 7 0 17 7 5 5
Others 0 0 0 0 0 0 0 0 0 0 0 0
Under 15K
15-24K
25-34K
35-49K
50-74K
75-99K
100-124K
125-149K
150-174K
175-199K
200-249K
250K+
0
10
20
30
40
50

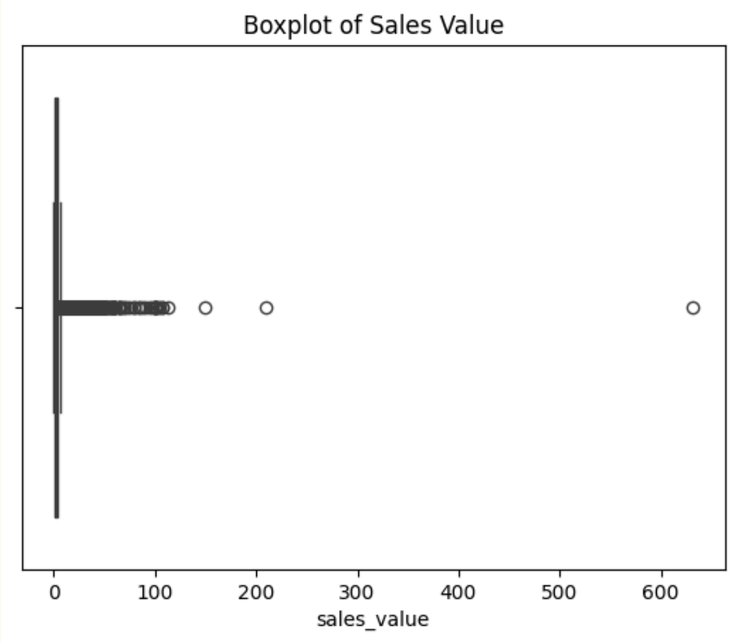

Brand Distribution
800000
600000
400000
200000
0
Count
National
Private
Brand


Comparison of Member Status Distribution
Dataset
Train
Validation
Test
Proportion (0 to 1)
0.0
0.1
0.2
0.3
0.4
0.5
0
1
Member Status

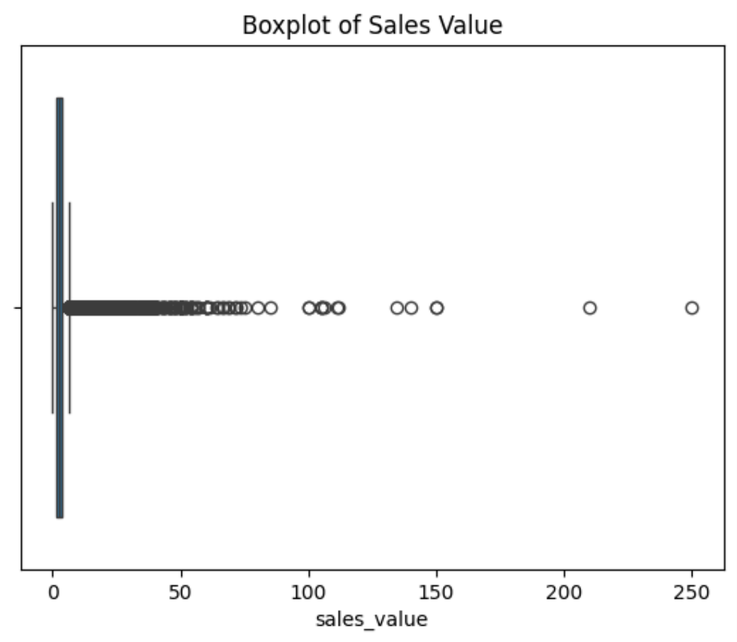

Comparison of Life Stage Distribution
Dataset
Train
Validation
Test
Proportion (0 to 1)
0.00
0.05
0.10
0.15
0.20
0.25
0
1
2
3
4
5
6
7
Life Stage



Train



Validation



Test